

Stanislav Smirnov

Cologne, Germany | +49 176 4345 8829 | mr.stassmirnoff@hotmail.com | linkedin.com/in/smirnov-stanislav | github.com/mrStasSmirnoff

SUMMARY

Senior Data Scientist focused on Payments, Risk and Optimizations. I lead ML and analytics initiatives across routing, retries, authorization and fraud detection, from problem framing and prioritization through production rollout and evaluation.

SKILLS

Leadership & Delivery: stakeholder management, cross-functional leadership, execution ownership, structured problem framing, technical mentoring, technical interviewing

Product & Strategy: KPI definition, prioritization, decision memos / PRDs, rollout planning, launch evaluation, roadmap/backlog definition, experimentation design

Payments & Fraud: authorization/decline analysis, PSP performance, routing, retries, tokenization (network tokens), payment optimization, fraud detection

Programming & ML/AI: Python, SQL, PySpark, classical ML, Deep Learning (CNNs, LSTMs)

Cloud & Data: AWS, Azure, Glue, Athena, Lambda, Data Vault 2.0, ETL development

Tools: Git, Docker, Airflow, FastAPI, SageMaker, MWAA, Databricks, TensorFlow, Keras

EXPERIENCE

Senior Data Scientist, Payments

April 2024 – Present

Cleverbridge AG – Merchant of Record and e-commerce payments platform

Cologne, DE

- Lead payment optimization and fraud ML initiatives across routing, retries, authorization, tokenization and fraud detection, from opportunity discovery and prioritization through stakeholder alignment, rollout, KPI definition and evaluation.
- Improved fallback re-routing authorization success by **4x** (from 2–3% to 11%) by identifying a weak route in North American traffic and driving the switch to a higher-performing PSP; resulted in a **3x** increase in recovered revenue on that traffic slice.
- Rebuilt and deployed a production fraud ML model with Fraud Prevention, revising the labelling logic and improving offline recall from **~44% to ~72%** at a **1% false-positive rate** (PR-AUC: **0.51 to 0.80**); benchmarked it against the existing model and an external vendor score for the build-vs-buy decision.
- Expanded **Dynamic Retries** to approximately **90% of the client base**, securing several hundred thousand euros in recovered revenue.
- Designed and implemented monitoring for client-level product KPIs and model drift
- Lead and mentor junior data scientists and contribute to technical hiring for DS and DE roles

Data Scientist

December 2021 – April 2024

Cleverbridge AG

Cologne, DE

- Proposed and launched **Dynamic Retries**, an ML-driven retry timing product; wrote the product proposal, aligned engineering, product, and business stakeholders, and led delivery from PoC through MVP to production rollout.
- Improved success rates for in-scope retries by **25–50%**, owning KPI definition, measurement design, and iteration based on post-launch performance.
- Designed a hybrid retry strategy combining a rule-based scheduling heuristic (country/currency × day-of-month × hour-of-day) with ML timing optimization to select high-conversion retry windows.
- Built reusable online and offline feature groups in AWS Glue and PySpark for rule-based retry strategies and ML training, and developed SageMaker pipelines for automated model training and inference.
- Designed the ML framework behind **CleverAutomations** and delivered churn-reduction models and automated workflows that identified at-risk customers and triggered targeted campaigns.

Data Engineer

September 2019 – December 2021

Cleverbridge AG

Cologne, DE

- Led the design and implementation of a scalable data warehouse on Azure (Data Vault 2.0), enabling reliable reporting and self-service analytics across the company.
- Built and optimized ETL pipelines (Azure Data Factory, Databricks, T-SQL).
- Developed Python microservices and Airflow workflows for production data products, using Docker and Flyway for maintainable services and versioned deployment.

Machine Learning Intern

May 2017 – July 2017

KOSTAL Group

Dortmund, DE

- Built a gesture-recognition CNN in Keras and TensorFlow, achieving **80%** validation accuracy on the top five gestures and improving robustness through preprocessing and data augmentation.

Network Optimization Engineer

May 2012 – March 2015

MegaFon

Republic of Bashkortostan, Russia

- Led the **network optimization team** of three engineers.
- Reduced CS call drop rate in the 2G network by **0.2%** and increased UMTS Inter-RAT handover success rate by **0.8%**

EDUCATION

University of Paderborn

Paderborn, DE

Master of Science in Electrical Systems Engineering (Signal Processing)

2015 – 2018

- **Thesis:** Enhancing Object Detection in Fine Art Paintings Using Deep Learning Techniques (+**18%** detection accuracy vs. baseline).
- **Research Assistant:** statistical image processing; plasmonic nanoparticles modeling in COMSOL; Lie groups programming in GAP.

PUBLICATIONS & CERTIFICATIONS

Deep Learning for Object Detection in Fine Art Paintings | *IEEE Xplore*

2020

Nonlinear Dielectric Properties of Random Paraelectric-Dielectric Composites | *Acta Materialia*

2021

Dynamic Retries: failed payment recovery | *Technical writing*

AWS Certified Cloud Practitioner

LANGUAGES

Russian (Native) | English (Fluent) | German (Upper Intermediate, B2)